



PORTFOLIO MANAGEMENT SYSTEM

A 3-Tier Web Application

The Portfolio Management System is a web-based application that centralizes security data, manages investment portfolios, monitors asset allocation, generates rebalancing alerts, and compares portfolio performance with the NIFTY 50 benchmark.



Hema Meena R
Jayashri K



Roles

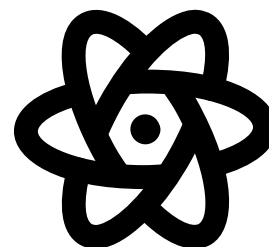
FUND MANAGER

- Login to the application
- View portfolio dashboard
- Search securities
- Create and activate portfolios
- Buy and sell securities
- View holdings and transactions
- Monitor asset allocation
- View alerts and rebalancing suggestions
- Compare portfolio return with NIFTY 50

SYSTEM

- Maintains Unified Security Master
- Fetches historical security prices
- Validates theme allocation limits
- Updates holdings after Buy/Sell
- Calculates portfolio value and P/L
- Generates alerts for deviation > 5%
- Suggests INCREASE / REDUCE / HOLD
- Calculates benchmark performance

FrontEnd



**React
and CSS**

BackEnd



**FastAPI
Python**

Database

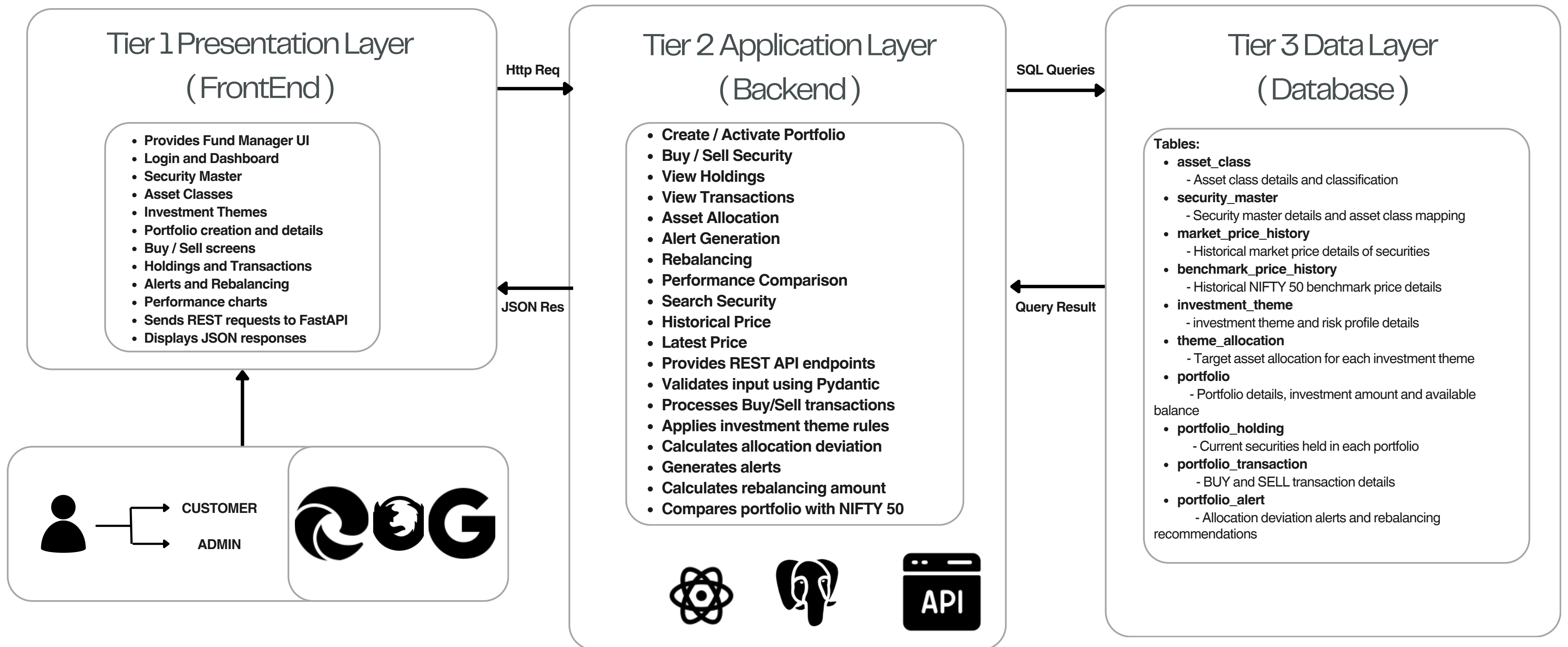


**Postgres
SQL**



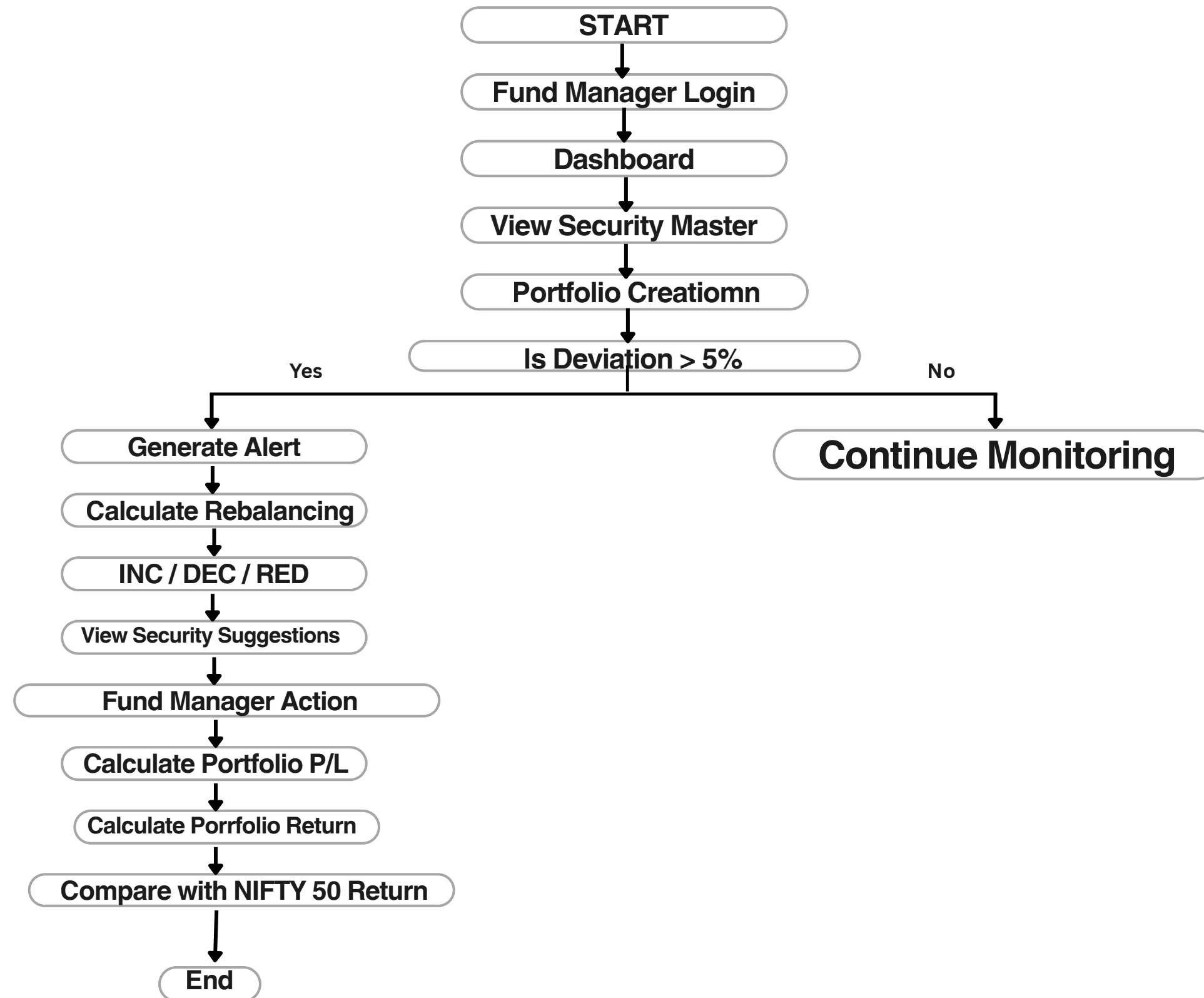


Architecture





Workflow





Implementation



CREATE

- Create Asset Class
- Create Investment Theme
- Create Portfolio
- Create Buy Transaction
- Create Sell Transaction
- Generate Portfolio Alerts



READ

- View Security Master
- View Portfolios , Charts
- View Holdings , Alerts
- View Transactions
- View Asset Allocation
- View Rebalancing Suggestions
- View Performance



UPDATE

- Update Asset Class
- Update Investment Theme
- Activate Portfolio
- Update Holdings after Buy/Sell
- Update Available Balance
- Mark Alerts as Read / Resolved
- Recalculate Allocation



DELETE

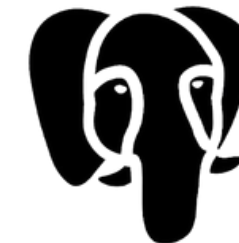
- Deactivate Asset Class
- Deactivate Investment Theme
- Remove holding after complete Sell
- Resolve alerts when deviation returns within limit



FRONTEND
(React + CSS)



BACKEND
(Python + FASTAPI)



DATABASE
(Postgres SQL)



API COMMUNICATION
(RestAPI + JSON)



VALIDATION
(Pydantic)

Thank You
